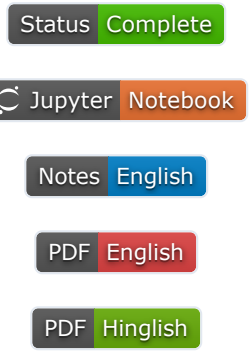
Lecture 30.04: Gradient Boosting Classifier Ka Python Code — Scikit-Learn Implementation [Hinglish]



📌 Topic Overview

Production implementation of GradientBoostingClassifier in Scikit-Learn. Covers dataset creation via make_classification, hyperparameter tuning, log_loss optimization, and comprehensive evaluation using accuracy, confusion matrices, precision, recall, and ROC-AUC.

📋 Is Topic Mein Hum Kya Seekhenge?

- Scikit-Learn se `GradientBoostingClassifier` import aur train karna.
- Loss function: `loss='log_loss'` (binary cross-entropy) ka significance.
- Classification evaluation: Accuracy, Precision, Recall, F1-Score, aur Confusion Matrix.
- Feature Importances (`feature_importances_`) ko inspect karna.
- Overfitting diagnose karna train vs test accuracy gap se.

🔍 Har Ek Point Ka Simple & Detailed Explanation

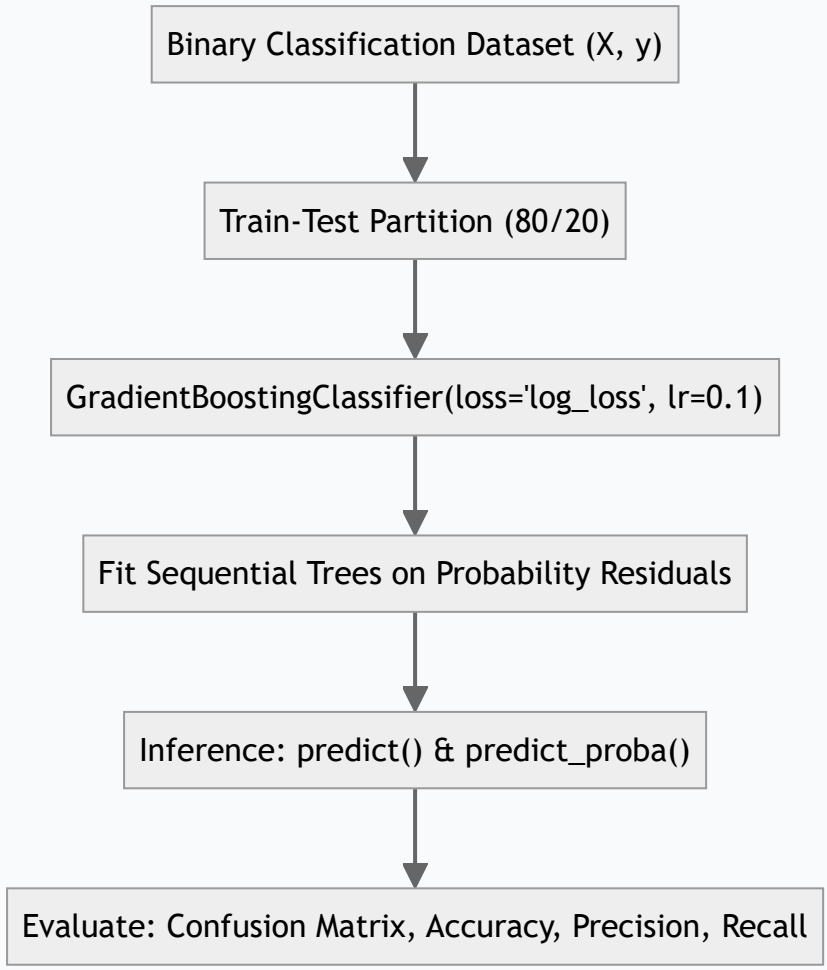
Scikit-Learn Classifier Setup: `from sklearn.ensemble import GradientBoostingClassifier` use karke classifier object banate hain.

Core Hyperparameters: `loss='log_loss'` , `n_estimators=100` , `learning_rate=0.1` , `max_depth=3` .

Feature Importance: GBC har feature ka Gini/MDI importance score calculate karta hai jo batata hai ki kis feature ne residuals ko sabse zyada reduce kiya.

Medical & Real-World Use Cases: Jahan classification accuracy ke sath-sath confidence probability zaroori hoti hai, wahan Gradient Boosting state-of-the-art results deta hai.

📊 Visual Flowchart / Diagram



🛠️ Mathematical Formulations

Binary cross-entropy loss (log_loss) optimized during boosting:

$$\mathcal{L}_{\log}(\mathbf{y}, \mathbf{p}) = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \ln(p_i) + (1-y_i)\ln(1-p_i)\right]$$

💻 Python Code Implementation

```
import numpy as np
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.ensemble import GradientBoostingClassifier
```

```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_scor

def train_gradient_boosting_classifier(n_samples=600, n_features=8, n_estimators=1
    """
    Builds, trains, and evaluates a Scikit-Learn GradientBoostingClassifier.
    """
    # 1. Synthetic binary classification dataset
    X, y = make_classification(
        n_samples=n_samples, n_features=n_features, n_informative=5, random_state=
    )
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42, stratify=y
    )

    # 2. Fit GradientBoostingClassifier
    gbc = GradientBoostingClassifier(
        loss='log_loss',
        learning_rate=learning_rate,
        n_estimators=n_estimators,
        max_depth=3,
        random_state=42
    )
    gbc.fit(X_train, y_train)

    # 3. Predict classes and probabilities
    y_pred = gbc.predict(X_test)
    y_prob = gbc.predict_proba(X_test)[:, 1]

    metrics = {
        "accuracy": float(accuracy_score(y_test, y_pred)),
        "precision": float(precision_score(y_test, y_pred, zero_division=0)),
        "recall": float(recall_score(y_test, y_pred, zero_division=0)),
        "f1_score": float(f1_score(y_test, y_pred, zero_division=0)),
        "confusion_matrix": confusion_matrix(y_test, y_pred).tolist()
    }

    return gbc, metrics

# ----------------- Driver Execution & Testing -----------------
if __name__ == "__main__":
    trained_gbc, perf = train_gradient_boosting_classifier()

    print("=" * 60)
    print("--- SCIKIT-LEARN GRADIENT BOOSTING CLASSIFIER PIPELINE ---")
    print(f"Accuracy  : {perf['accuracy'] * 100:.2f}%")
    print(f"Precision : {perf['precision'] * 100:.2f}%")
    print(f"Recall    : {perf['recall'] * 100:.2f}%")
    print(f"F1-Score  : {perf['f1_score'] * 100:.2f}%")
    print(f"Confusion Matrix: {perf['confusion_matrix']}")
    print("=" * 60)
```

📌 Summary Points & Key Takeaways

1. **Sequential Learning**: Har agla model pichhle model ki galtiyon ko seekhta hai.
2. **Shrinkage (Learning Rate)**: 0.1 jaise chote kadam lene se model overfit hone se bachta hai.
3. **Weak Learners**: Boosting mein shallow decision trees (depth 1 se 3) use kiye jaate hain.